

Labo 27.02 - Calculs

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Objectif trouver les résistances

Thevenin Equivalents

$V_{TH} = \frac{R_2}{R_1 + R_2} V_1$
 $V_{TH} = x V_1$
 $x = \frac{R_2}{R_1 + R_2}$
 $0 \leq x \leq 1$
 $R_{TH} = \frac{R_1 R_2}{R_1 + R_2} = x R_1$

$V_{TH} = \frac{R_2}{R_1 + R_2} V_1 + \frac{R_1}{R_1 + R_2} V_2$
 $V_{TH} = x V_1 + (1 - x) V_2$
 $x = \frac{R_2}{R_1 + R_2}$
 $0 \leq x \leq 1$
 $R_{TH} = \frac{R_1 R_2}{R_1 + R_2} = x R_1$

Si 0, est à la masse, si 1 on a V1 et pas le pont diviseur

Non-inverting Comparator

$V_{TH1} = \frac{R_2}{R_1 + R_2} V_{in}$
 $x = \frac{R_2}{R_1 + R_2}$
 $V_{TH1} = x V_{in}$
 $R_{TH1} = \frac{R_1 R_2}{R_1 + R_2}$
 $R_{TH1} = x R_1$
 $V_{TH2} = \frac{R_5}{R_4 + R_5} V_{ref}$
 $y = \frac{R_5}{R_4 + R_5}$
 $V_{TH2} = y V_{ref}$

$V_{TH1} = x V_{in}$
 $V_{TH2} = y V_{ref}$

Non-inverting Comparator

$V_{TH3} = \frac{R_3}{x R_1 + R_3} x V_{in} + \frac{x R_1}{x R_1 + R_3} V_{out}$
 $z = \frac{R_3}{x R_1 + R_3}$
 $V_{TH3} = x z V_{in} + (1 - z) V_{out}$

Level transitions when:

$v_+ = v_-$
 $y V_{ref} = x z V_{in} + (1 - z) V_{out}$
 $x = 1 \text{ or } y = 1$

$V_{out} = V_{in}$ *passé à* $V_{out} = V_{ref}$

$$x = 1 \text{ or } y = 1$$

$$V_{out} = 0V \xrightarrow{\text{passé à}} V_{inH}$$

$$yV_{ref} = xzV_{inH}$$

$$z = \frac{y}{x} \frac{V_{ref}}{V_{inH}}$$

$$V_{out} = V_{DD} \xrightarrow{\text{passé à}} V_{inL}$$

$$yV_{ref} = xzV_{inL} + (1-z)V_{DD}$$

Non-inverting Comparator

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$xyV_{ref}V_{inH} = xyV_{ref}V_{inL} + xV_{inH}V_{DD} - yV_{ref}V_{DD}$

$y = 1$

$xV_{ref}V_{inH} = xV_{ref}V_{inL} + xV_{inH}V_{DD} - V_{ref}V_{DD}$

$V_{ref}V_{DD} = x(V_{ref}V_{inL} + V_{inH}V_{DD} - V_{ref}V_{inH})$

$$x = \frac{V_{ref}V_{DD}}{V_{inH}V_{DD} - V_{ref}V_{inH} + V_{ref}V_{inL}}$$

$$z = \frac{1}{x} \frac{V_{ref}}{V_{inH}}$$

$x = 1$

$yV_{ref}V_{inH} = yV_{ref}V_{inL} + V_{inH}V_{DD} - yV_{ref}V_{DD}$

$y(V_{ref}V_{inH} + V_{ref}V_{DD} - V_{ref}V_{inL}) = V_{inH}V_{DD}$

$$y = \frac{V_{inH}V_{DD}}{V_{ref}V_{DD} + V_{ref}V_{inH} - V_{ref}V_{inL}}$$

$$z = y \frac{V_{ref}}{V_{inH}}$$

Non-inverting Comparator

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$z = \frac{R_3}{xR_1 + R_3}$

$x = \frac{R_2}{R_1 + R_2}$

$y = \frac{R_5}{R_4 + R_5}$

R_3 : choice

$z(xR_1 + R_3) = R_3$

$R_1 = \frac{1-z}{xz} R_3$

If $y < 1$

R_5 : choice

$R_4 = \frac{1-y}{y} R_5$

If $x < 1$

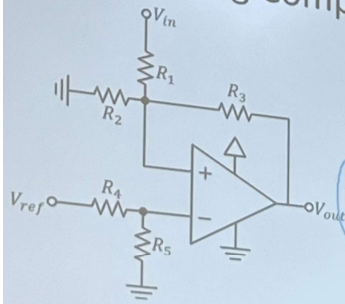
$xR_1 = R_2(1-x)$

$R_2 = \frac{x}{1-x} R_1$

$V_{inH} = V_{ref} \frac{y}{xz}$

$V_{inL} = \frac{yV_{ref} - (1-z)V_{DD}}{xz}$

Non-inverting Comparator



- Known: $V_{DD}, V_{ref}, V_{inH}, V_{inL}$
1. Calculate x, y and z
 2. Calculate R_1, R_2, R_3, R_4, R_5
 3. Calculate the *sensitivity* for each of the resistors:
When R changes by 1% how much does V_{inH} and V_{inL} change?
 4. Normalize the resistors
 5. Recalculate V_{inH} and V_{inL}
 6. Do we need to normalize certain